Equations

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

Product

$$\prod_{k=1}^{n} k = n!$$

Integral

$$\int_{0}^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

Matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Limit

$$\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^{x} = e$$

Cases

$$f(x) = \begin{cases} x^2 & x \geq 0 \\ -x & x < 0 \end{cases}$$

Equation array

$$\begin{aligned} x + y &= 1 \\ x - y &= 3 \end{aligned}$$

Accents and bar

$$\overline{(x+y)} = x \, \hat{} \, + y \, \tilde{}$$

Box

$$\boxed{a = b}$$

Functions

$$\sin^2\theta + \cos^2\theta = 1$$

Roots

$$\sqrt[3]{x} + \sqrt[3]{8} = 2 + \sqrt{x^2 + y^2}$$

Einstein's relation $E = mc^2$ sits inside this sentence, between ordinary words.